

Safety Data Sheet

According to Notification of Thailand's Ministry of Industry B.E. 2555 (2012)



Petronas Hydraulic 100

Version 0.0

Release Date: 2017/05/08

1. PRODUCT AND COMPANY IDENTIFICATION

Product name : Petronas Hydraulic 100
Other means of identification : None
Recommended use : Lubricant for hydraulic system
Limitation of use : This product should not be used for purposes other than those specified without the advice of an expert
Supplier : PETRONAS International Marketing (Thailand) Co., Ltd
Registered corporate address : 388 Exchange Tower, Unit Nos. 4101-4104 41st floor, Sukhumvit Road, Kweang Klongtoey, Khet Klongtoey, Bangkok 10110
Emergency phone numbers : +662-663 5881-5

Manufacturer or supplier's details

2. HAZARDS IDENTIFICATION

GHS Classification

Acute aquatic toxicity : Category 3
Chronic aquatic toxicity : Category 3

GHS Label element

Hazard pictograms : None
Signal word : None
Hazard statements : H412 Harmful to aquatic life with long lasting effects.
Precautionary statements : **Prevention:**
P273 Avoid release to the environment.
Disposal:
P501 Dispose of contents/ container to an approved waste disposal plant.

Other hazards which do not result in classification

No information available.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Substance / Mixture : Mixture

Hazardous components

Chemical Name	CAS-No.	Concentration (%)
Paraffinic Distillate, Heavy, Hydrotreated	64742-54-7	≥ 90 - ≤ 100
Alkyl Phenol	128-39-2	≥ 0.1 - < 1

4. FIRST AID MEASURES

If inhaled : If unconscious, place in recovery position and seek medical advice.
If symptoms persist, call a medical doctor.
In case of eye contact : Remove contact lenses.
Protect unharmed eye.

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- If swallowed : If eye irritation persists, consult a specialist.
: Keep respiratory tract clear.
Do NOT give milk or alcoholic beverages.
Never give anything by mouth to an unconscious person.
If symptoms persist, call a medical doctor.
- General advice : Do not leave the victim unattended.
- Most important symptoms and effects, both acute and delayed : No information available.

5. FIREFIGHTING MEASURES

- Specific hazards during firefighting : Do not allow run-off from fire fighting to enter drains or water courses.
- Specific extinguishing methods : Collect contaminated fire extinguishing water separately. This must not be discharged into drains.
Fire residues and contaminated fire extinguishing water must be disposed off in accordance with local regulations.
- Special protective equipment for firefighters : Wear self-contained breathing apparatus for firefighting if necessary.

6. ACCIDENTAL RELEASE MEASURES

- Environmental precautions : If the product contaminates rivers and lakes or drains inform respective authorities.
Prevent product from entering drains.
- Methods and materials for containment and cleaning up : Wipe up with absorbent material (e.g. cloth, fleece).
Keep in suitable, closed containers for disposal.

7. HANDLING AND STORAGE

- Advice on protection against fire and explosion : Normal measures for preventive fire protection.
- Advice on safe handling : For personal protection see section 8.
Smoking, eating and drinking should be prohibited in the application area.
- Conditions for safe storage : Containers which are opened must be carefully resealed and kept upright to prevent leakage.
Electrical installations / working materials must comply with the technological safety standards.
- Materials to avoid : No materials to be especially mentioned.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Components with workplace control parameters

Components	CAS-No.	Value type (Form of exposure)	Control parameters / Permissible concentration	Basis
Paraffinic Distillate, Heavy, Hydrotreated	64742-54-7	TWA (Inhalable fraction)	5 mg/m ³	ACGIH

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	Further information: Upper Respiratory Tract irritation, Not classifiable as a human carcinogen
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Personal protective equipment

Respiratory protection	: No personal respiratory protective equipment normally required.
Eye protection	: Safety glasses
Skin and body protection	: Protective suit
Hygiene measures	: General industrial hygiene practice.

9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance	: Liquid
Flash point	: 218 °C Method: open cup
Viscosity, kinematic	: 11.19 mm ² /s (100 °C) 100.12 mm ² /s (40 °C)

10. STABILITY AND REACTIVITY

Reactivity	: Hazardous polymerisation does not occur.
Chemical stability	: Stable under normal conditions.
Possibility of hazardous reactions	: No dangerous reaction known under conditions of normal use.
Conditions to avoid	: Heat, sparks, flame and build-up of static electricity.
Incompatible materials	: Strong oxidizing agents.
Hazardous decomposition products	: Fumes, smoke, carbon monoxide

11. TOXICOLOGICAL INFORMATION

Acute toxicity

Components:

Paraffinic Distillate, Heavy, Hydrotreated:

Acute oral toxicity	: LD50 (Rat): > 5,000 mg/kg
Acute inhalation toxicity	: LC50 (Rat): > 5.53 mg/l
Acute dermal toxicity	: LD50 (Rabbit): > 2,000 mg/kg

Alkyl Phenol:

Acute oral toxicity	: Remarks: No data available
Acute inhalation toxicity	: Remarks: No data available
Acute dermal toxicity	: Remarks: No data available

Skin corrosion/irritation

Components:

Paraffinic Distillate, Heavy, Hydrotreated:

Species	: Rabbit
Result	: No skin irritation

Alkyl Phenol:

Result	: Skin irritation
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Serious eye damage/eye irritation

Components:

Paraffinic Distillate, Heavy, Hydrotreated:

Species : Rabbit
Result : No eye irritation

Respiratory or skin sensitisation

Components:

Paraffinic Distillate, Heavy, Hydrotreated:

Exposure routes : Skin contact
Result : No evidence of skin sensitization.
Exposure routes : Inhalation
Remarks : No data available

Alkyl Phenol:

Remarks : No data available

Germ cell mutagenicity

Components:

Paraffinic Distillate, Heavy, Hydrotreated:

Germ cell mutagenicity - : No data available
Assessment

Alkyl Phenol:

Germ cell mutagenicity - : No data available
Assessment

Carcinogenicity

Components:

Paraffinic Distillate, Heavy, Hydrotreated:

Carcinogenicity - : Contains less than 3 % DMSO extract as measured by IP 346
Assessment 'Determination of polycyclic aromatics in unused lubricating base oils and asphaltene free petroleum fractions — Dimethyl sulphoxide extraction refractive index method'

Alkyl Phenol:

Carcinogenicity - : No data available
Assessment

Reproductive toxicity

Components:

Paraffinic Distillate, Heavy, Hydrotreated:

Reproductive toxicity - : Not suspected to be a reproductive toxicant.
Assessment

Alkyl Phenol:

Reproductive toxicity - : No data available
Assessment

STOT - single exposure

Components:

Paraffinic Distillate, Heavy, Hydrotreated:

Assessment : No specific organ toxicity following single exposure.

Alkyl Phenol:

Remarks : No data available

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STOT - repeated exposure

Components:

Paraffinic Distillate, Heavy, Hydrotreated:

Assessment : No specific organ toxicity following repeated exposure.

Alkyl Phenol:

Remarks : No data available

12. ECOLOGICAL INFORMATION

Ecotoxicity

Components:

Paraffinic Distillate, Heavy, Hydrotreated:

Toxicity to fish : (Fish): > 100 mg/l
Toxicity to daphnia and other aquatic invertebrates : (Crustaceans): > 100 mg/l
Toxicity to algae : (Algae): > 100 mg/l
Toxicity to fish (Chronic toxicity) : > 1 mg/l
Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : > 1 mg/l
Toxicity to microorganisms : Remarks: No data available

Alkyl Phenol:

Toxicity to daphnia and other aquatic invertebrates : Remarks: No data available
Toxicity to fish (Chronic toxicity) : Chronic Toxicity Value (Fish): 0.006 mg/l
Exposure time: 30 d
Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : Chronic Toxicity Value (Daphnia (water flea)): 0.080 mg/l
Exposure time: 21 d
Toxicity to microorganisms : Remarks: No data available
Ecotoxicology Assessment
Acute aquatic toxicity : Very toxic to aquatic life.

Chronic aquatic toxicity : Very toxic to aquatic life with long lasting effects.

Persistence and degradability

Components:

Paraffinic Distillate, Heavy, Hydrotreated:

Biodegradability : Result: Inherently biodegradable.

Alkyl Phenol:

Biodegradability : Result: Not readily biodegradable.

Bioaccumulative potential

Components:

Paraffinic Distillate, Heavy, Hydrotreated:

Bioaccumulation : Remarks: Potentially bio-accumulative.
Partition coefficient: n-octanol/water : log Pow: 2 - 6

Alkyl Phenol:

Bioaccumulation : Bioconcentration factor (BCF): 2.69

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Partition coefficient: n-octanol/water : log Pow: 4.48

Mobility in soil

Components:

Paraffinic Distillate, Heavy, Hydrotreated:

Mobility : Medium: Soil
Remarks: No data available

Other adverse effects

Product:

Additional ecological information : An environmental hazard cannot be excluded in the event of unprofessional handling or disposal.
Harmful to aquatic life with long lasting effects.

Components:

Paraffinic Distillate, Heavy, Hydrotreated:

Additional ecological information : No data available

Alkyl Phenol:No data available

13. DISPOSAL CONSIDERATIONS

Disposal methods

Waste from residues : The product should not be allowed to enter drains, water courses or the soil.

14. TRANSPORT INFORMATION

International Regulation

UNRTDG

Not regulated as a dangerous good

IATA-DGR

Not regulated as a dangerous good

IMDG-Code

Not regulated as a dangerous good

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not applicable for product as supplied.

15. REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

The components of this product are reported in the following inventories:

CH INV : Not in compliance with the inventory
TSCA : Not On TSCA Inventory
DSL : This product contains the following components that are not on the Canadian DSL nor NDSL.

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	Zinc alkyldithiophosphate	
	Mineral Oil	
	Acrylic copolymer	
	Calcium Sulfonate	
	Alkylated Phenol	121158-58-5
	Substituted triazole	
	Alkylated phenol	
AICS	: Not in compliance with the inventory	
NZIoC	: Not in compliance with the inventory	
ENCS	: Not in compliance with the inventory	
ISHL	: Not in compliance with the inventory	
KECI	: Not in compliance with the inventory	
PICCS	: Not in compliance with the inventory	
IECSC	: Not in compliance with the inventory	

16. OTHER INFORMATION

Revision Date : 2017/05/08

Full text of other abbreviations

(Q)SAR	-	(Quantitative) Structure Activity Relationship
ACGIH	-	American Conference of Governmental Industrial Hygienists
AICS	-	Australian Inventory of Chemical Substances
ANTT	-	National Agency for Transport by Land of Brazil
ASTM	-	American Society for the Testing of Materials
bw	-	Body weight
CCHC	-	Chemicals Classification and Hazard Communication
CMR	-	Carcinogen, Mutagen or Reproductive Toxicant
CPR	-	Controlled Products Regulations
DIN	-	Standard of the German Institute for Standardisation
DSL	-	Domestic Substances List (Canada)
ECx	-	Concentration associated with x% response
ELx	-	Loading rate associated with x% response
EmS	-	Emergency Schedule
ENCS	-	Existing and New Chemical Substances (Japan)
ErCx	-	Concentration associated with x% growth rate response
ERG	-	Emergency Response Guide
GHS	-	Globally Harmonized System
GLP	-	Good Laboratory Practice
IARC	-	International Agency for Research on Cancer
IATA	-	International Air Transport Association
IBC	-	International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk
IC50	-	Half maximal inhibitory concentration
ICAO	-	International Civil Aviation Organization
ICOP	-	Industry Code of Practice on Chemicals Classification and Hazard Communication
IECSC	-	Inventory of Existing Chemical Substances in China
IMDG	-	International Maritime Dangerous Goods
IMO	-	International Maritime Organization
ISHL	-	Industrial Safety and Health Law (Japan)
ISO	-	International Organisation for Standardization
KECI	-	Korea Existing Chemicals Inventory
LC50	-	Lethal Concentration to 50 % of a test population
LD50	-	Lethal Dose to 50% of a test population (Median Lethal Dose)
MARPOL	-	International Convention for the Prevention of Pollution from Ships

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MY PEL	-	Malaysian Permissible Exposure Limit
n.o.s.	-	Not Otherwise Specified
Nch	-	Chilean Norm
NITE	-	National Institute of Technology and Evaluation
NO(A)EC	-	No Observed (Adverse) Effect Concentration
NO(A)EL	-	No Observed (Adverse) Effect Level
NOELR	-	No Observable Effect Loading Rate
NOM	-	Official Mexican Norm
NTP	-	National Toxicology Program
NZIoC	-	New Zealand Inventory of Chemicals
OCSPP	-	Office of Chemical Safety and Pollution Prevention
OECD	-	Organization for Economic Co-operation and Development
PBT	-	Persistent, Bioaccumulative and Toxic
PICCS	-	Philippines Inventory of Chemicals and Chemical Substances
REACH	-	Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals
SADT	-	Self-Accelerating Decomposition Temperature
SDS	-	Safety Data Sheet
STEL	-	Short Term Exposure Limit
TCSI	-	Taiwan Chemical Substance Inventory
TDG	-	Transportation of Dangerous Goods
TSCA	-	Toxic Substances Control Act (United States)
TWA	-	Time Weighted Average
UN	-	United Nations
UNRTDG	-	United Nations Recommendations on the Transport of Dangerous Goods
UVCB	-	Unknown or Variable Composition, Complex Reaction Products and Biological Materials
vPvB	-	Very Persistent and Very Bioaccumulative
WHMIS	-	Workplace Hazardous Materials Information System

Disclaimer

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Product Stewardship Advisory:

PETRONAS aims to increase awareness of all the hazards associated with the storage, handling and use of its products. Thoroughly reviewing the accompanying Safety Data Sheets and disseminating the information to all dependent and interested parties is an essential part of any 'Responsible Care' programme.

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